

NAMAN YESHWANTH KUMAR

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SUMMARY

Forward-deployed-by-default full-stack engineer (MS Computer Engineering, ASU — GPA 3.81, May 2026) who translates customer workflows into shipped agentic systems. Sole engineer for 18 months at NCR GOLD — sat with one customer, reverse-engineered their legacy ERP (no public API), built 3 role-based portals + an end-user WhatsApp chatbot that automated their entire order lifecycle. Solo-shipped Trezzit (app.trezzit.com), a production AI platform serving **600+ users** with a 21-tool MCP server, multi-provider LLM routing, and a pipeline iterated from ~60% → 95%+ accuracy.

TECHNICAL SKILLS

Full-Stack & TypeScript: Python | TypeScript | React 19 | Next.js | Node.js | Material-UI | Tailwind | Django REST Framework | FastAPI | PostgreSQL | Redis | Celery | WebSocket | PWA

AI Agents & LLMs: LangGraph | LangChain | ReAct Agents | MCP Protocol | Tool Use / Function Calling | Multi-provider LLM Routing (Gemini / OpenAI / Claude / Grok) | RAG Pipelines | Pydantic Validation | Cursor | Claude Code

Cloud & Customer-Facing Delivery: Reverse-engineering legacy systems without API | WhatsApp Business API | MQTT / Kafka | AWS | Docker | DigitalOcean | Vercel | Supabase | Nginx | GitHub Actions CI/CD | Linux | Git

PROJECT EXPERIENCE

Trezzit — Production AI Platform with Agentic Workflows, Sole Engineer (app.trezzit.com)

Feb 2025 – Present

- Solo-built and shipped Trezzit (app.trezzit.com), a production AI platform serving **600+ users** for 12+ months — React 19 + Django + Python with **341 components, 60+ REST endpoints, 40 DB models, 995 tests**; PostgreSQL (Supabase), Redis, Celery + Celery Beat for async background jobs; PWA with iOS Safari compat, VAPID push, Share Target API.
- Built a two-stage AI receipt pipeline routing across Gemini, OpenAI, Claude, and Grok — pass 1 + semantic validation gate + automatic model-upgrade fallback on failure — iterated parsing accuracy from ~60% → 95%+; pre-computed balance system with O(1) reads, zero-sum invariant enforcement, and optimistic locking on edits.
- Built an **MCP server** letting AI assistants take actions across the platform (create bills, query balances, upload receipts) over the same auth and data layer as the web app; production observability tracks per-endpoint p95 latency and per-provider LLM tokens / cost.

SiliconCrew — 21-Tool LangGraph Agent Architecture with MCP Server

Nov 2025 – Present

- Built a LangGraph ReAct agent (21 tools, multi-provider routing Gemini / OpenAI / Anthropic) that drives an end-to-end workflow from natural language input to a verified, deployed artifact — self-corrects on failure (read diagnostic → identify root cause → patch → re-verify); full MCP server (stdio / SSE / Streamable HTTP) compatible with Claude Desktop and any MCP client; achieved **37/92 on CVDP**; public at github.com/naman-ranka/siliconcrew.

HEAL.AI — Insurance-Policy RAG & Automated Dispute Generation (Devlabs Hackathon)

Sep – Nov 2025

- Built HEAL.AI at Devlabs Hackathon — custom RAG pipeline (PDF chunking + 768-dim embeddings + cosine-similarity retrieval) over **uploaded insurance policies** enabling natural-language Q&A, billing overcharge detection, and **automated dispute-email generation** via Gemini 2.5; FastAPI + Next.js + TypeScript, Dockerized — same unstructured-doc + workflow-automation pattern as enterprise insurance AI.

Intel Automated Self-Checkout — Open Source Contribution

Jan – Feb 2025

- Contributed sensor analytics microservice to **Intel's official self-checkout reference architecture** — configurable data pipeline (MQTT, Kafka, HTTP) for real-time LiDAR and weight sensor data with Grafana dashboard; **PR #655 merged upstream** after Intel engineering review.

WORK EXPERIENCE

Software Developer (Sole Engineer)

Oct 2022 – Apr 2024

NCR GOLD

Bengaluru, India

- Embedded with the customer for 18 months as **sole engineer** — translated their actual order-management workflow into a production Django platform (PostgreSQL + Celery + Nginx on DigitalOcean) processing **18,000+ orders over 3 years**; built 3 role-based portals (admin, artisan, customer) with multi-method auth and automated PDF invoice + WhatsApp Business API delivery on every state transition, eliminating ~20 min of manual relay per order.
- Reverse-engineered third-party Jewely ERP (**zero public API**) via direct SQL schema analysis and built a bidirectional bridge model syncing production weights and ERP status in real time; engineered an end-user-facing WhatsApp state-machine chatbot (JSON state per user, admin + customer flows) — shipped the platform that runs their entire order business without disrupting the legacy system.

EDUCATION

Master of Science — Computer Engineering (GPA 3.81/4.0)

Expected May 2026

Arizona State University — Coursework: Artificial Intelligence · Foundations of Algorithms · Reinforcement Learning in Robotics · Perception in Robotics · Algorithm/Hardware Co-Design

Tempe, AZ

Bachelor of Engineering — Electrical and Electronics

August 2022

BMS College of Engineering

Bengaluru, India

AWARDS & HONORS

- Won **1st place at Devlabs Hackathon** with BillBusters (team), then **placed 2nd at the subsequent Devlabs Hackathon** solo-rebuilding it as HEAL.AI — insurance-policy RAG with automated dispute-email generation.
- Won **'Best Use of AI'** at Strategy X DevHacks'25 (ASU) — built AdaptED AI, a Gemini-powered platform generating personalized flashcards, quizzes, and learning paths from uploaded documents in a 24-hour hackathon.
- Secured **2nd place at Hack SoDA 2024** (Amazon-sponsored, 24-hour) — built PassGen, a secure offline Chrome Extension for unique password generation.